

Periodic Bivariate-Bicycle Autoresearch

Internal progress report for the 2026-08-06-periodic-bb campaign

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Preliminary, local-only results. Nothing in this report has been submitted to the public board. The reported distance labels are witness-backed upper bounds; no distance lower bound has yet been certified. Exact MILP certification is pending, literature/state-of-the-art novelty has not yet been investigated, and independent external reproduction has not yet been performed.

Abstract

We report progress in a continuous automated search over periodic rectangular bivariate-bicycle (BB) quantum LDPC codes. The run is orchestrated by OpenAI GPT-5.6 Sol through the Codex harness and follows the Unitary Foundation qLDPC Challenge’s tool-agnostic autoresearch protocol (“autoresearch mode” here): generate reproducible constructions, screen exact structural properties, persist every logical witness, attack promoted candidates at increasing randomized-search budgets, and let the repository’s trusted validator—rather than the agent—decide whether a result is a campaign find. The campaign may be viewed as a continuation of our earlier LLM-guided BB-code search, with a different harness and repository trust gate [1]. At the stated data cutoff, 47,163 structural proposals have produced 45 board-advancing local finds representing 44 distinct parameter triples. Two distinct $[[696, 236, d \leq 3]]$ constructions jointly maximize the campaign score $\mathcal{S} = k^2 d_w / n$ at 240.07. The largest witness-supported distance label among the finds is $d \leq 9$, attained by six constructions. All distance statements remain upper bounds and all novelty statements remain provisional.

1 Scope, terminology, and status

The construction family is locked to the periodic abelian group $G = \mathbb{Z}_\ell \times \mathbb{Z}_m$. Open-boundary, twisted-torus, non-abelian generalized-bicycle, coset, lifted-product, and balanced-product codes are out of scope. Work remains on the local `autoresearch/2026-08-06-local` branch (based on public fork of the challenge <https://github.com/cbjuan/qldpc-challenge>); no candidate has been written to `codes/`, pushed, submitted, or used to open a pull request. The surrounding verification and leaderboard framework is described in the QEC Challenge whitepaper [2]; the current public board is the challenge website [3].

Data cutoff and inclusion rule. This is a frozen working-tree snapshot at **2026-08-07 16:02:20 CEST**, based on repository commit `ef7f13b` plus local artifacts already present at that time. Git committedness and scientific validation are separate properties. A candidate enters the results only when a frozen candidate JSON and its adjacent full verdict both exist by the cutoff, with `passed:true`

and `novelty.board_advancing:true`. Under that rule, 41 finds are committed at the cutoff and four additional finds (c45014–c45017) are uncommitted but fully verdict-backed; all 45 are included. Candidate files without a trusted verdict are excluded from counts, rankings, and extrema regardless of their screening weight or Git status. Files written after the cutoff are likewise outside this report.

The campaign uses the following deliberately narrow meaning of a *find*: the saved call to `verify/validate_candidate.py` returns `passed:true`, and its novelty gate returns `board_advancing:true`. “Board-advancing” is relative only to the repository snapshot and its computed Pareto cells. It is not a claim of literature novelty or of global state-of-the-art performance.

For compactness, this report writes a campaign label as $\llbracket n, k, d \leq d_w \rrbracket$. Here d_w is the lightest logical weight persisted by the campaign, so the mathematical statement is

$$d_X \leq w_X, \quad d_Z \leq w_Z, \quad d \leq d_w := \min(w_X, w_Z).$$

Thus $\llbracket n, k, d \leq d_w \rrbracket$ must not be read as an exact-distance certificate. The same caveat applies to every score containing d_w .

Table 1: Campaign snapshot. Counts of validator outcomes are by distinct candidate ID, not by verdict file; three candidates were validated at more than one confirmation rung.

Quantity	Count
Deterministic structural proposals	47,163
Unique encoding proposals retained in the structural ledger	18,004
Canonical construction duplicates skipped	8,776
Known translated-equal structural traps skipped	140
Candidate IDs given a trusted-validator verdict	83
Candidate IDs passing the validator	81
Board-advancing campaign finds	45
committed at the cutoff	41
uncommitted but verdict-backed at the cutoff	4
Distinct $\llbracket n, k, d \leq d_w \rrbracket$ triples among the finds	44
Active promotion-queue candidates without a verdict	64
Public submissions	0

Of the 46 entries retained in the findings ledger at the cutoff, 20 reached a one-million accelerated-trial rung, 20 reached a 60,000-trial rung, and six are shallow 2,000-trial or calibration results. Forty-five are finds; the remaining $\llbracket 40, 2, d \leq 4 \rrbracket$ entry is a passed but dominated calibration code.

2 Construction and exact structural screening

Let

$$R = \mathbb{F}_2[x, y]/(x^\ell - 1, y^m - 1),$$

and let $A, B \in R$ be arbitrary binary sums of monomials, with unequal support sizes allowed. Their regular representations are $\ell m \times \ell m$ binary circulant matrices. The CSS checks are

$$H_X = \begin{bmatrix} A & B \end{bmatrix}, \quad H_Z = \begin{bmatrix} B^\top & A^\top \end{bmatrix}.$$

Because A and B commute and the arithmetic has characteristic two, $H_X H_Z^T = AB + BA = 0$. The resulting block length and encoded dimension are recomputed exactly as

$$n = 2\ell m, \quad k = n - \text{rank}_{\mathbb{F}_2}(H_X) - \text{rank}_{\mathbb{F}_2}(H_Z).$$

The search respects the repository bounds $n \leq 700$ and total check weight at most 32.

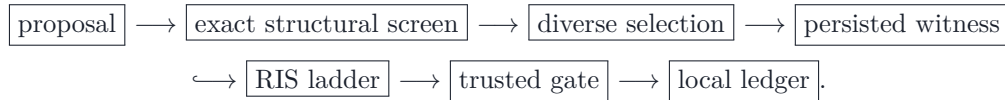
Before any distance work, the campaign performs exact GF(2) rank and CSS checks, canonicalizes translations and axis symmetries, removes construction duplicates, records connected-component structure, and rejects a known translated- $A = B$ trap that forces a weight-two logical whenever the code encodes. This distance-free census makes the expensive witness search a promotion step rather than a property of every proposal.

3 Automation and research workflow

The two automation layers have different responsibilities.

- **Codex harness.** GPT-5.6 Sol operates as a repository-aware research agent: it inspects the current board and field notes, implements and invokes the deterministic campaign runner, monitors artifacts, adjusts allocation according to observed collapses, and maintains the decision journal. Shell execution, file persistence, and source-control awareness make long multi-rung experiments resumable and auditable.
- **Unitary Foundation autoresearch protocol (“autoresearch mode”).** The repository’s tool-agnostic autoresearch manual supplies the scientific control loop and trust boundary. The model may propose, rank, and test constructions, but it cannot declare its own result valid. Every find must be packaged by the research kit, preserve its witnesses, and pass the pinned validator. The trusted `verify/` stack is never modified by the campaign.

The implemented loop is



Proposal IDs and random seeds are deterministic. The initial lanes cover weight-4 breadth, high-rate weight-6 and weight-8 gaps, arbitrary unequal supports, axis-mixed supports, shared-factor motifs, and mutations of reproducible BB seeds. Later phases add targeted nearby-geometry mutations, controlled-order common-factor transfers, and a high-distance continuation around parents carrying reported bounds $d \leq 8$, $d \leq 9$, $d \leq 14$, or $d \leq 18$. A parent’s reported distance label is a selection prior only: every child obtains and persists its own witness.

The key operational safeguard is persistence-first execution. `submit.make_submission` packages an initial Python result; every deeper accelerated rung saves both raw witness evidence and a new candidate file without overwriting its ancestor. Accelerated witnesses are then rechecked by the trusted Python GF(2) predicates. Consequently, an expensive logical operator is not lost if a later run or save fails.

4 Verification already performed and still pending

Table 2: Verification layers and their present campaign status.

Layer	Method	Status and interpretation
Structure	Exact binary rank, CSS commutation, dimensions, check weight, schema and resource bounds	Complete for packaged candidates; these are deterministic checks.
Logical witness	Exact tests that an X (Z) vector lies in $\ker H_Z \setminus \text{row } H_X$ (respectively $\ker H_X \setminus \text{row } H_Z$), with exact weight	Complete and embedded for both sectors; proves only an upper bound.
Distance attack	Randomized-information-set (RIS/QDistRnd-style) search using random-permuted elimination and light-row combinations; NumPy first, then the bit-packed <code>gf2_fast</code> implementation at 8k, 60k, or 1M rungs	Complete at the recorded budget. Python and C++ are implementations of the same search family, not independent distance algorithms.
Trusted candidate gate	Fresh-seed pure-Python bounded RIS refutation, exact witness recheck, stabilizer fingerprint deduplication, and board-Pareto labeling	Complete for all ledger entries. A pass means “not refuted within this budget”, not “distance proven”.
Exact distance	<code>research/kit/distance.exact_distance</code> , wrapping the SciPy/HiGHS MILP tier in <code>verify/certify.py</code> , separately for X and Z	Pending. No MILP certificate or <code>d_exact</code> artifact is present in the campaign.
Independent heuristic	BP+OSD residual-logical search via <code>research/kit/distance.decoder_distance</code>	Available but not yet run. It would provide an independent upper-bound cross-check, not an exact proof.
External validation	Independent reproduction and literature/state-of-the-art comparison	Not started.

The pending exact tier formulates a binary mixed-integer linear program for each logical sector and each complementary logical-basis generator. It asks whether a nontrivial logical of weight $1, \dots, d_w - 1$ exists while enforcing the parity-check constraints and an odd logical overlap. Infeasibility of every such problem on both sectors certifies the persisted upper bound as the exact distance; a feasible solution refutes it; a timeout leaves the result at upper-bound confidence. The first MILP queue should prioritize the six $d \leq 9$ finds and the six $d \leq 8$ finds, while retaining all timeout and counterexample artifacts.

5 Results

The campaign tracks

$$\mathcal{S} = \frac{k^2 d_w}{n} \quad \text{and} \quad \mathcal{B} = \frac{k d_w^2}{n}.$$

$\mathcal{S}$ is the campaign's rate-sensitive priority requested for this run; $\mathcal{B}$ is the board's commonly reported operational-efficiency column. Neither scalar replaces Pareto comparison. Because d_w is an upper bound, the displayed values are provisional upper-bound scores as well.

5.1 Highlights

Table 3: Leading candidates under complementary provisional views.

Criterion	Candidate(s)	Value	Interpretation
Largest $\mathcal{S}$	c40000, c40002: [[696, 236, $d \leq 3$]]	240.07	Two distinct saved constructions, on $\mathbb{Z}_6 \times \mathbb{Z}_{58}$ and $\mathbb{Z}_{12} \times \mathbb{Z}_{29}$; the gain is rate-driven.
Runner-up $\mathcal{S}$	c40003: [[690, 234, $d \leq 3$]]	238.07	Another order-three, high-rate continuation.
Largest reported upper bound	c10524, c20283, c45014–c45017: six $m = 15$ constructions	$d \leq 9$	The same 4+2 support motif on six torus geometries; exact distance remains pending for every entry.
Largest $\mathcal{B}$	c20118: [[272, 48, $d \leq 8$]]	11.29	Best balance of rate and the witnessed upper-bound label under $k d_w^2 / n$.

The two $\mathcal{S}$ leaders have equal parameters but are not the same saved construction: their canonical construction keys and torus geometries differ. Each candidate's gate found neither an exact board duplicate nor a Weisfeiler–Lehman-equivalent board entry. This establishes two independently board-advancing constructions, not inequivalence under every possible qubit permutation.

The leading constructions are directly reproducible from their supports:

$$\begin{aligned}
\text{c40000 : } & (\ell, m) = (6, 58), \\
& A = \{(0, 0), (1, 25), (3, 25), (4, 0)\}, \\
& B = \{(0, 0), (1, 12), (3, 12), (4, 0)\}; \\
\text{c40002 : } & (\ell, m) = (12, 29), \\
& A = \{(1, 25), (4, 0), (5, 25), (8, 0)\}, \\
& B = \{(1, 12), (4, 0), (5, 12), (8, 0)\}; \\
\text{c10524/c20283/c45014–c45017 : } & m = 15, \quad \ell \in \{9, 11, 19, 20, 22, 23\}, \\
& A = \{(0, 0), (0, 11), (7, 2), (7, 8)\}, \quad B = \{(0, 0), (0, 1)\}.
\end{aligned}$$

The campaign runner calls the last four candidates “exact geometry transfers.” In that provenance label, “exact” means that the integer support was transferred without mutation; it does *not* mean that the quantum distance has been certified exactly.

5.2 Complete inventory of campaign finds

Table 4 lists every board-advancing candidate once, ordered by $\mathcal{S}$. It contains 45 constructions but 44 parameter triples because the first two rows are distinct $\llbracket 696, 236, d \leq 3 \rrbracket$ codes. The check-weight distribution is 24 weight-6, 20 weight-8, and one weight-16 find. The witnessed upper-bound distribution is: three at $d \leq 2$, twelve at $d \leq 3$, six at $d \leq 4$, eight at $d \leq 6$, four at $d \leq 7$, six at $d \leq 8$, and six at $d \leq 9$.

Table 4: Complete find inventory. w is the maximum check weight; all entries passed the trusted gate and were labeled board-advancing. Purple marks the joint provisional $\mathcal{S} = k^2 d_w/n$ leaders; green marks the six constructions attaining the campaign’s largest witness-supported label, $d \leq 9$. Both highlights depend on witness upper bounds and do not denote exact-distance certificates.

ID	Campaign label	w	$\mathcal{S} = k^2 d_w/n$	$\mathcal{B} = k d_w^2/n$	Note
c40000	$\llbracket 696, 236, d \leq 3 \rrbracket$	8	240.07	3.05	joint $\mathcal{S}$ leader
c40002	$\llbracket 696, 236, d \leq 3 \rrbracket$	8	240.07	3.05	joint $\mathcal{S}$ leader; distinct
c40003	$\llbracket 690, 234, d \leq 3 \rrbracket$	8	238.07	3.05	$\mathcal{S}$ runner-up
c40004	$\llbracket 672, 228, d \leq 3 \rrbracket$	8	232.07	3.05	—
c40001	$\llbracket 696, 232, d \leq 3 \rrbracket$	6	232.00	3.00	weight-6 high-rate leader
c4417	$\llbracket 452, 228, d \leq 2 \rrbracket$	8	230.02	2.02	$d \leq 2$; audit-only priority
c40005	$\llbracket 696, 174, d \leq 4 \rrbracket$	8	174.00	4.00	—
c40006	$\llbracket 688, 172, d \leq 4 \rrbracket$	8	172.00	4.00	—
c40007	$\llbracket 680, 170, d \leq 4 \rrbracket$	8	170.00	4.00	—
c14474	$\llbracket 396, 136, d \leq 3 \rrbracket$	8	140.12	3.09	—
c12783	$\llbracket 224, 112, d \leq 2 \rrbracket$	6	112.00	2.00	$d \leq 2$; audit-only priority
c16394	$\llbracket 288, 100, d \leq 3 \rrbracket$	8	104.17	3.13	—
c19513	$\llbracket 384, 96, d \leq 4 \rrbracket$	8	96.00	4.00	—
c16039	$\llbracket 288, 96, d \leq 3 \rrbracket$	8	96.00	3.00	—
c11462	$\llbracket 270, 90, d \leq 3 \rrbracket$	6	90.00	3.00	—
c6423	$\llbracket 234, 78, d \leq 3 \rrbracket$	6	78.00	3.00	—
c20118	$\llbracket 272, 48, d \leq 8 \rrbracket$	8	67.76	11.29	largest $\mathcal{B}$
c20058	$\llbracket 304, 52, d \leq 7 \rrbracket$	8	62.26	8.38	—
c19531	$\llbracket 248, 62, d \leq 4 \rrbracket$	6	62.00	4.00	—
c20978	$\llbracket 266, 50, d \leq 6 \rrbracket$	8	56.39	6.77	—
c45591	$\llbracket 448, 56, d \leq 8 \rrbracket$	8	56.00	8.00	—
c20343	$\llbracket 252, 48, d \leq 6 \rrbracket$	8	54.86	6.86	—
c20198	$\llbracket 238, 46, d \leq 6 \rrbracket$	8	53.34	6.96	—
c20498	$\llbracket 224, 44, d \leq 6 \rrbracket$	8	51.86	7.07	—
c20718	$\llbracket 256, 46, d \leq 6 \rrbracket$	8	49.59	6.47	locally dominated by c20198

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Table 4 (continued)

ID	Campaign label	w	$\mathcal{S}$	$\mathcal{B}$	Note
c2359	$\llbracket 108, 36, d \leq 3 \rrbracket$	16	36.00	3.00	weight-9plus cell
c45014	$\llbracket 690, 46, d \leq 9 \rrbracket$	6	27.60	5.40	local; verdict-backed
c45015	$\llbracket 660, 44, d \leq 9 \rrbracket$	6	26.40	5.40	local; verdict-backed
c46163	$\llbracket 644, 46, d \leq 8 \rrbracket$	6	26.29	4.57	—
c142	$\llbracket 208, 52, d \leq 2 \rrbracket$	6	26.00	1.00	$d \leq 2$; audit-only priority
c46164	$\llbracket 616, 44, d \leq 8 \rrbracket$	6	25.14	4.57	—
c45016	$\llbracket 600, 40, d \leq 9 \rrbracket$	6	24.00	5.40	local; verdict-backed
c20418	$\llbracket 270, 30, d \leq 7 \rrbracket$	6	23.33	5.44	—
c45017	$\llbracket 570, 38, d \leq 9 \rrbracket$	6	22.80	5.40	local; verdict-backed
c20458	$\llbracket 240, 30, d \leq 6 \rrbracket$	6	22.50	4.50	—
c46587	$\llbracket 532, 38, d \leq 8 \rrbracket$	6	21.71	4.57	—
c6625	$\llbracket 266, 38, d \leq 4 \rrbracket$	6	21.71	2.29	—
c20918	$\llbracket 234, 26, d \leq 7 \rrbracket$	6	20.22	5.44	—
c16229	$\llbracket 208, 26, d \leq 6 \rrbracket$	6	19.50	4.50	—
c20623	$\llbracket 260, 26, d \leq 6 \rrbracket$	6	15.60	3.60	locally dominated by c16229
c20283	$\llbracket 330, 22, d \leq 9 \rrbracket$	6	13.20	5.40	joint largest bound $d \leq 9$
c20168	$\llbracket 280, 20, d \leq 8 \rrbracket$	6	11.43	4.57	—
c10524	$\llbracket 270, 18, d \leq 9 \rrbracket$	6	10.80	5.40	joint largest bound $d \leq 9$
c4584	$\llbracket 198, 26, d \leq 3 \rrbracket$	6	10.24	1.18	—
c18628	$\llbracket 280, 20, d \leq 7 \rrbracket$	6	10.00	3.50	—

The additional calibration entry c8081, $\llbracket 40, 2, d \leq 4 \rrbracket$ with $w = 4$, has $\mathcal{S} = 0.40$ and $\mathcal{B} = 0.80$. It passed structural, witness, and bounded-refutation checks but was dominated by the current board, so it is not counted as a find.

5.3 Negative results and high-distance controls

The trusted gate was run on 83 distinct candidate IDs. Besides the 45 finds, 36 candidates passed but did not advance their board cell. The largest witnessed upper-bound label anywhere in this accepted control set is $d \leq 24$, attained by c45020, $\llbracket 432, 12, d \leq 24 \rrbracket$; it is excluded from the find inventory because it is non-advancing. Two additional candidates did not pass solely because the gate identified exact existing-board duplicates: $\llbracket 288, 50, d \leq 8 \rrbracket$ and $\llbracket 288, 12, d \leq 18 \rrbracket$. These controls are useful for calibrating the high-distance continuation, but they are not new campaign finds.

5.4 In-flight candidates excluded from the results

At the data cutoff, 64 additional packaged candidates remained in three active survivor queues without any saved `validate_candidate.py` verdict: 17 had reached 60,000 accelerated RIS trials, 23 had reached 8,000, and 24 had only the initial 400-trial Python screen. None is a campaign find, and none contributes to the inventory, distributions, rankings, or extrema above. Their persisted logical weights are screening upper bounds only and may decrease under deeper attacks; the largest

such shallow label was $d \leq 54$, which is especially susceptible to search-depth inflation. The strings `exact-grid` and `exact-rate` in two queue filenames describe deterministic proposal lanes, not exact-distance certification. Later artifacts written after the cutoff are deferred to a subsequent report revision.

6 Limitations and immediate next steps

1. **External reproduction.** Rebuild the highlighted candidates from their monomial supports in a clean checkout and on an independent machine, rerun fresh-seed witness searches, and compare file hashes and validator verdicts. This is the immediate next validation step.
2. **Exact MILP certification.** Run the SciPy/HiGHS certifier on both logical sectors, beginning with the $d \leq 9$ and $d \leq 8$ finds. Save feasible counterexamples, logged infeasibility outcomes, and all timeouts; do not relabel a distance exact unless the authoritative certifier returns `d_exact: true`.
3. **Independent distance corroboration.** Run the BP+OSD residual-witness method with fresh seeds. Agreement would be valuable cross-mechanism evidence, but would still not supply a lower bound.
4. **State-of-the-art and equivalence review.** No systematic comparison with the current literature, preprints, or independent code tables has begun. Search for equivalent BB support presentations and qubit permutations, compare against published parameter families, and retain the label “literature novelty unverified” until that review is complete.
5. **Human promotion decision.** Only after reproduction, certification/corroboration, and literature review should a human decide whether any staged JSON belongs in `codes/`. This report itself authorizes no submission.

The numerical source of record for the inventory is `FINDINGS.md` together with the adjacent candidate and `*.verdict.json` artifacts. Search policy and reproducible commands are recorded in `README.md`; the deterministic implementation is `campaign.py`; and the governing protocol is `research/AUTORESEARCH.md`. This report is a dated snapshot of an ongoing local campaign.

References

- [1] J. Cruz-Benito, A. W. Cross, D. Kremer, and I. Faro, “Evolutionary Discovery of Bivariate Bicycle Codes with LLM-Guided Search,” *arXiv preprint arXiv:2606.02418*, 2026. [arXiv:2606.02418](#).
- [2] Unitary Foundation, “The QEC Challenge: A public, automatically verified leaderboard for quantum low-density parity-check codes,” 2026. [QEC Challenge whitepaper](#).
- [3] Unitary Foundation, “qLDPC Challenge—public leaderboard,” 2026. [qLDPC Challenge website](#).

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As stated above, these are preliminary results and external reproduction is the immediate next validation step.